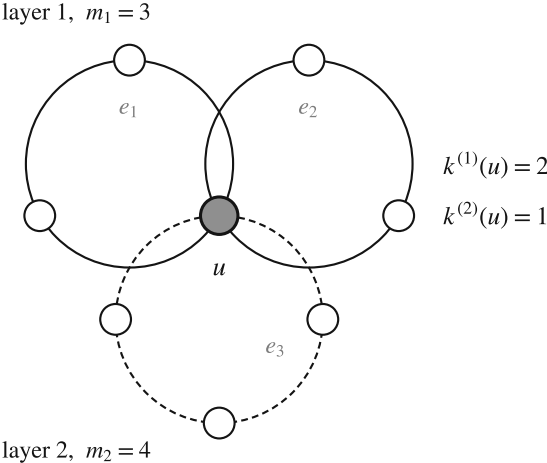
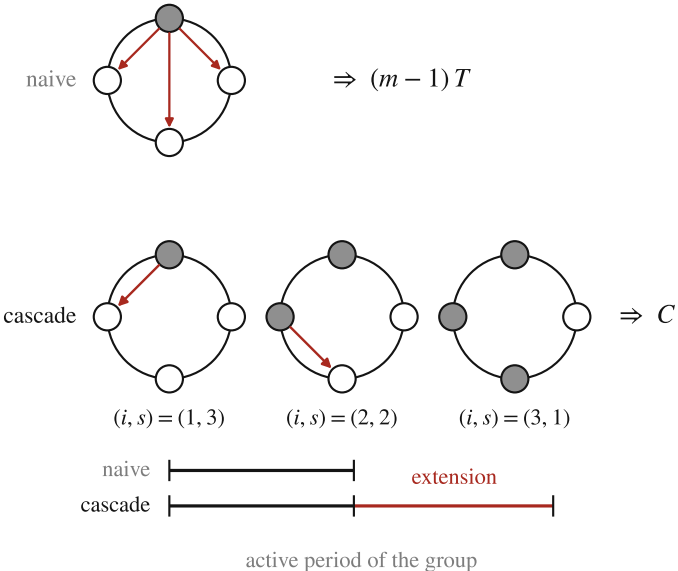


a. multiplex hypergraph

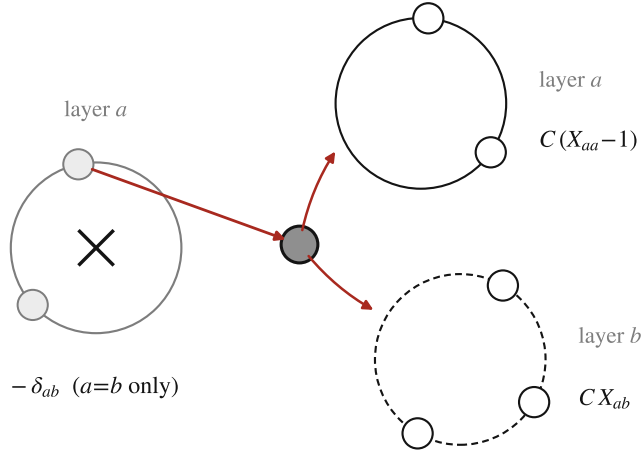


b. intra-group cascade



$$C(m, \lambda, \theta) > (m - 1) T$$

c. group-level branching



$$N_{ab} = C(X_{ab} - \delta_{ab})$$